

Project Initialization and Planning Phase

Date	28 June 2026
Team ID	
Project Title	Rising Waters
Maximum Marks	5 Marks

Project Overview

Field	Details
Objective	Develop a machine learning-based flood prediction system that analyzes historical weather and rainfall parameters to predict flood risk and support early disaster preparedness.
Scope	Includes data preprocessing, exploratory data analysis, model training, evaluation, and deployment of the best-performing Random Forest model through a Flask web application.

Problem Statement

Field	Details
Description	Manual flood analysis is time-consuming and may delay critical decisions during heavy rainfall. An intelligent prediction system is required for faster flood risk assessment.
Impact	Early flood prediction improves disaster preparedness, supports evacuation planning, reduces property damage, and helps protect lives.

Proposed Solution

Field	Details
Approach	Train and deploy a Random Forest classifier using historical weather and rainfall data. User inputs are preprocessed with StandardScaler before prediction.
Key Features	• Random Forest prediction model • Flask web application • Responsive user interface • Instant Flood / No Flood prediction • Input validation • Modular architecture

Resource Requirements

Resource Type	Description	Specification / Allocation
Hardware	Development System	Intel i5/Ryzen 5, 8 GB RAM, 20 GB Storage
Software	Frameworks & Libraries	Flask, Scikit-learn, Pandas, NumPy, Matplotlib, Seaborn, Joblib
Development	IDE	Visual Studio Code, Jupyter Notebook
Data	Dataset	Historical Weather & Rainfall Dataset (CSV)